

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 5: Scientific Names and Binomial Nomenclature: One Name for Every Organism

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	D	A	A	D	A	D

USE AFTER: PREDICT PHASE — Predict: individual prediction and Think-Pair-Share on problems that arise when people use only their own local names

Q1. A farmer in Kisumu and a researcher in Nairobi each use only their own local name for a mosquito species when sharing information about malaria control. What problem is most likely to happen?

- A. Local names are always wrong, so neither person can describe the mosquito
- B. The mosquito will change its behaviour depending on the name used
- C. They may not be sure they are talking about the same organism, so mistakes can happen**
- D. The mosquito can only be studied by people who speak Kiswahili

Correct answer: C. Common names vary by place and language, so two people may use different names for the same organism, or the same name for different ones. This causes confusion in farming, medicine and pest control.

USE AFTER: OBSERVE PHASE — Part A: Name Comparison Table, examples where one common name refers to different organisms

Q2. In some countries the word 'corn' means maize, but in other countries it means a different cereal. What does this show about common names?

- A. A common name always tells us exactly which species an organism is
- B. Common names are only used for plants, never for animals
- C. Common names are the same everywhere in the world
- D. The same common name can refer to different organisms, which causes confusion**

Correct answer: D. One common name can refer to different organisms in different places. This is one reason a universal name is needed.

USE AFTER: OBSERVE PHASE — Part C: Jigsaw, expert rule 3 (italics or underlining each part separately)

Q3. You want to handwrite the scientific name of the goat, *Capra hircus*, following the rules of binomial nomenclature. What should you do?

- A. Underline the genus and the species separately**
- B. Write the whole name in capital letters instead of underlining
- C. Underline the whole name with one continuous line so the two words are joined
- D. Write the name in brackets so that it can be told apart from other words

Correct answer: A. Scientific names are written in italics when typed. When handwritten, the genus and the species are each underlined separately.

USE AFTER: EXPLAIN PHASE — Card-sorting task: sorting 'Correct' and 'Incorrect' name cards and explaining which rule was broken

Q4. A learner writes the scientific name of the mosquito as *Anopheles Gambiae*. Which rule of binomial nomenclature has been broken?

- A. The species name must start with a small letter**
- B. The genus must start with a small letter
- C. A scientific name must be written in Kiswahili
- D. A scientific name must have three parts

Correct answer: A. The genus begins with a capital letter and the species begins with a small letter. The correct form is *Anopheles gambiae*.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning (CER) explanation: 'Why do scientists need a scientific name for each organism?'

Q5. A group writes the claim: 'Scientists need a scientific name for each organism even though people already have common names.' Which reasoning best supports this claim?

- A. Scientific names are longer than common names, so they are more useful
- B. Scientific names are in Latin, and Latin is the easiest language for everyone to speak
- C. Local names are wrong and should not be used by anyone
- D. Each organism has one accepted two-part name used worldwide, so scientists everywhere know which organism is meant and can share information without confusion**

Correct answer: D. Good reasoning links the evidence (common names differ, but *Zea mays* is the same everywhere) to the idea that a universal name avoids confusion.

USE AFTER: EXPLAIN PHASE — Class discussion and debate: 'Should we stop using local names now that we have scientific names?'

Q6. In a class debate, a learner says: 'Now that we have scientific names, we should stop using local names like mbuzi, mbūri and diel.' Which response is best?

- A. Disagree, because local names remain valuable in communities, while scientific names are used for communication among scientists worldwide**
- B. Agree, because scientific names replace local names completely
- C. Disagree, because scientific names can only be used for plants
- D. Agree, because local names are never correct

Correct answer: A. Scientific names do not replace local names. Local names keep their cultural and community value, and scientific names give one shared name for universal scientific communication.

USE AFTER: MODEL BUILDING PHASE — Revising the class classification model: Naming Layer diagram labelling genus and species

Q7. Learners add a Naming Layer to the class model and label the scientific name of the goat, *Capra hircus*. Which labelling is correct?

- A. Genus: *Capra hircus*; Species: goat
- B. Genus: mbuzi; Species: *Capra hircus*
- C. Genus: hircus; Species: *Capra*
- D. Genus: *Capra*; Species: hircus**

Correct answer: D. The first word, written with a capital letter, is the genus (*Capra*). The second word, written with a small letter, is the species (*hircus*).